COUPLED DICTIONARY LEARNING AND FEATURE MAPPING FOR CROSS-MODAL RETRIEVAL

Xing Xu, Atsushi Shimada, Rin-ichiro Taniguchi

Kyushu University, Japan
{xing, atsushi}@limu.ait.kyushu-u.ac.jp, rin@kyudai.jp

Li He

Qualcomm R&D Center, USA
lih@qti.qualcomm.com

ABSTRACT

In this paper, we investigate the problem of modeling images and associated text for cross-modal retrieval tasks such as text-to-image search and image-to-text search. To make the data from image and text modalities comparable, previous cross-modal retrieval methods directly learn two projection matrices to map the raw features of the two modalities into a common subspace, in which cross-modal data matching can be performed. However, the different feature representations and correlation structures of different modalities inhibit these methods from efficiently modeling the relationships across modalities through a common subspace. To handle the diversities of different modalities, we first leverage the coupled dictionary learning method to generate homogeneous sparse representations for different modalities by associating and jointly updating their dictionaries. We then use a coupled feature mapping scheme to project the derived sparse representations from different modalities into a common subspace in which cross-modal retrieval can be performed. Experiments on a variety of cross-modal retrieval tasks demonstrate that the proposed method outperforms the state-of-the-art approaches.

Index Terms— Cross-modal retrieval, coupled dictionary learning, feature mapping, image annotation

1. INTRODUCTION

During the past decade, there has been a massive explosion of multimedia content such as images and videos coming from multiple input channels on the Internet. Specifically, Internet images are usually associated with text. Figure 1 illustrates concrete examples of the Internet images with text. Usually, the associated text are used to describe the semantic content of these images. In particular, the term "*keywords*" refers to high level labels or categories, and the terms "*text*" and "*tags*" are textual descriptions taking from the surrounding webpages and the user-provided tags respectively. It is worth noting that the semantics of an image may be given by one or more *keywords* and the *text/tags* are typically noisy and may not explicitly mention the groundtruth *keywords*.

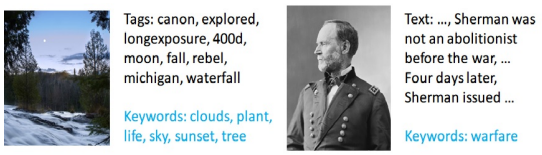


Fig. 1. Examples of Internet images with associated text.

Recently, jointly modeling the Internet images and associated text has been an active research area, resulting in a variety of cross-modal retrieval scenarios: (1) "*Text2Img*": automatically associating one or more semantically relevant images given a piece of text (i.e., a label, phrase or caption) [1, 2, 3], (2) "*Img2Text*": automatically annotating images using semantic labels [4, 5, 6] or describing images with phrases [3]. Due to the distinct statistical properties from the different modalities of image and text, one of the primary issues involved in the cross-modal retrieval is learning useful representations about such data by fusing the modalities into a joint representation. Indeed, the modalities of image and text may have different kinds of representations and correlational structures. For example, an image is usually represented using real-value and dense feature descriptors, whereas text is represented as discrete sparse word count vectors. However, many existing methods only consider the direct correlation between the original representations of images and corresponding text, leading to inefficient latent subspaces for representing data of both modalities. Moreover, these methods also have difficulties in discovering the highly non-linear relationship across different modalities.

In this work, we focus on learning a more efficient latent subspace from the original representations of different modalities for improved cross-modal retrieval results and propose a unified framework (see Fig. 2). The proposed framework consists of *coupled dictionary learning* and *coupled feature mapping*. We first transform the multimodal data into sparse representations via coupled dictionary learning, with the guarantee that the generated representations are homogeneous. We then map these representations using coupled linear re-

This research has been partly supported by Grant-in-Aid for Scientific Research (B), Grant Number 24300074.

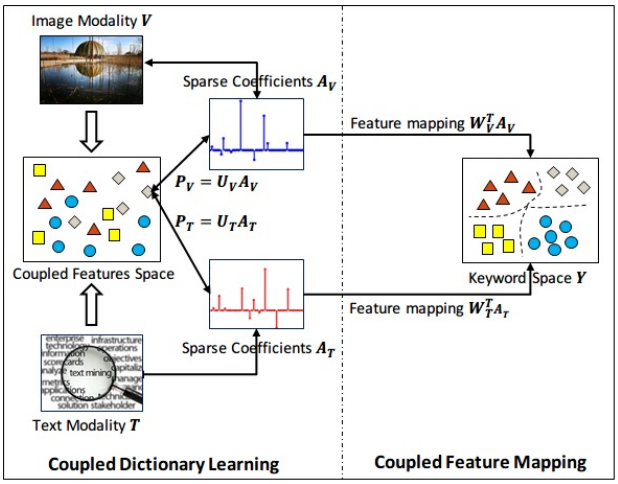


Fig. 2. Illustration of our proposed framework for cross-modal retrieval. Note that $\{\mathbf{A}_V, \mathbf{A}_T\}$ and $\{\mathbf{W}_V, \mathbf{W}_T\}$ are the sparse representations and projection matrices for image and text modalities, respectively.

gression where data from each modality are simultaneously projected into a common subspace of keywords. In test stage (i.e. Img2Text or Text2Img), we first generate the sparse representations for the given query from one modality using the learned coupled dictionary of this modality, then map the sparse representations into the common (keyword) subspace using the learned projection matrix. Finally, in the common subspace, we choose the best match from the other modality as the output.

The main contributions of our work can be summarized as follows: 1) We propose a novel framework that unifies coupled dictionary learning and coupled feature mapping for the cross-modal retrieval problem. The proposed framework can not only handle the diversity of different modal data, but also improve the efficiency of subspace learning and coupled feature selection. 2) We extend the sparse representation method of coupled dictionary learning developed for single modality to the case of multi-modality data, and derive an efficient iterative algorithm to solve the complex minimization problem. 3) We evaluate the proposed framework with a variety of cross-modal retrieval tasks on three challenging datasets, and the experimental results show that our proposed framework outperforms several relevant state-of-the-art approaches.

2. RELATED WORK

The problems of image and text retrieval are well-studied research topics [7]. Some of the earliest research are based on retrieval of unimodal data, i.e., both query and retrieved data belong to the same modality (either image or text). For example, given a text query, the unimodal methods directly match the query with textual meta-data on the web instead of examining the corresponding image. Usually, these unimodal approaches cannot be applied to cross-modal retrieval.

Recently, a number of successful approaches [6, 8, 9, 10] designed for cross-modal retrieval are proposed to jointly model images and associated text. The major goal of these works is learning a common latent space for image and text modalities, in which both modalities are projected into the same space for cross-modal retrieval. Specifically, several approaches rely on Canonical Correlation Analysis (CCA), a classic technique that maps two views, such as visual and textual features, into a common latent space where the correlation between the two views is maximized. Hwang and Grauman [10] have modeled the relative importance of words based on the order from the user provided annotations to improve cross-modal retrieval results. Ballan *et al.* [6] have used kernel CCA (KCCA) to develop cross-view approach to associate images and text. Gong *et al.* [9] have leveraged the CCA for cross-modal retrieval under the multi-view learning framework, in which the image and text views are linked to the underlying third view of semantic labels.

Except for the CCA based approaches, there are some other methods for the cross-modal retrieval. Chen *et al.* [11] have applied the Partial Least Squares (PLS) to cross-modal document retrieval. They use PLS to transform the visual features into the text space, then learn a semantic space to measure the similarity between two different modalities. Recently, Sharma *et al.* [12] made a comprehensive analysis for the multi-view learning framework to deal with cross-modal retrieval problem. They extend traditional discriminative methods, i.e., Linear Discriminant Analysis (LDA), Marginal Fisher Analysis (MFA), to the multi-view counterpart: Generalized Multi-view LDA (GMLDA) and Generalized Multi-view MFA (GMMFA). Wang *et al.* [13] proposed a generic framework to jointly perform common subspace learning and coupled feature selection from different modalities of data. They unified coupled linear regression, L_{21} norm and trace norm regularization terms into the generic framework and achieved the state-of-the-art performance for cross-media retrieval task.

We can learn from these methods that learning the common latent space is crucial for matching data of different modalities. However, when learning the latent space, these methods only considered the direct correlation between image and text modalities, ignoring the intrinsic diversity of representations and correlational structures in them. Therefore, these methods have difficulty in incorporating the highly nonlinear relationship between the low-level features across different modalities. To overcome this, we take advantage of dictionary learning to obtain the sparse representations of different modalities in a coupled learning manner. The sparse representations are homogenous for different modalities and can incorporate the relationship across modalities, ensuring learning a more representative latent space. The details of our proposed framework for cross-modal retrieval will be described in the next section.

3. PROPOSED FRAMEWORK

We first present the details of the two procedures: coupled dictionary learning and coupled feature mapping as shown in Fig. 2 and then develop iterative algorithm for learning parameters in each procedure. Finally, we describe the test phase of the proposed framework to cross-modal retrieval.

3.1. Coupled dictionary learning

Let $\mathcal{D} = \{I_1, ..., I_n\}$ be a collection of n images, each image $I_i = (\mathbf{v}_i, \mathbf{t}_i)$ consists of features from two modalities: d_1 dimensional visual feature vector and d_2 dimensional textual feature vector. Here, we denote data from image modality as $\mathbf{V} = [\mathbf{v}_1, ..., \mathbf{v}_n] \in \mathbb{R}^{d_1 \times n}$ and data from text modality as $\mathbf{T} = [\mathbf{t}_1, ..., \mathbf{t}_n] \in \mathbb{R}^{d_2 \times n}$, respectively. To handle the diversity of different modalities $\mathbf{V}$ and $\mathbf{T}$, we use sparse representation from dictionary learning for each modality since it has been shown to be very effective in data representation or reconstruction tasks. It is worth noting that the dictionary learning for different modalities is coupled and we impose associate function $f(\cdot)$ to relate the sparse representations of different modalities. The coupled dictionary learning can be formulated as the following minimization problem:

$$\begin{aligned} \min_{\mathbf{D}_V, \mathbf{D}_T, \mathbf{A}_V, \mathbf{A}_T} & \|\mathbf{V} - \mathbf{D}_V \mathbf{A}_V\|_F^2 + \|\mathbf{T} - \mathbf{D}_T \mathbf{A}_T\|_F^2 \\ & + \sigma(\|\mathbf{A}_V\|_1 + \|\mathbf{A}_T\|_1) + f(\mathbf{A}_V, \mathbf{A}_T) \quad (1) \\ & s.t. \quad \|\mathbf{d}_{v,i}\|_2 \leq 1, \|\mathbf{d}_{t,i}\|_2 \leq 1, \forall i. \end{aligned}$$

In Eq. 1, the original features $\mathbf{V}$ and $\mathbf{T}$ of the two modalities are reconstructed from the sparse representations $\mathbf{A}_V \in \mathbb{R}^{k_1 \times n}$ and $\mathbf{A}_T \in \mathbb{R}^{k_2 \times n}$, respectively. Here $\mathbf{A}_V$ and $\mathbf{A}_T$ rely on the learnt dictionaries $\mathbf{D}_V \in \mathbb{R}^{d_1 \times k_1}$ and $\mathbf{D}_T \in \mathbb{R}^{d_2 \times k_2}$, where k_1 and k_2 are dictionary size for $\mathbf{D}_V$ and $\mathbf{D}_T$, respectively. And $f(\mathbf{A}_V, \mathbf{A}_T)$ is the associating function defining the cross-modal relationship in terms of $\mathbf{A}_V$ and $\mathbf{A}_T$. Once the relationship between $\mathbf{A}_V$ and $\mathbf{A}_T$ is captured, the dictionaries $\mathbf{D}_V$ and $\mathbf{D}_T$ can be updated accordingly.

Regarding $f(\mathbf{A}_V, \mathbf{A}_T)$, following the scheme in [14], we introduce a k_c dimensional common feature space $\mathcal{P}$ to relate the sparse representations of different modalities. Here we restrict $k_c = k_1 = k_2$ so that different modalities of data should be comparable in $\mathcal{P}$. In addition, $\mathbf{P}_V = \mathbf{U}_V \mathbf{A}_V$ is the projected data from sparse representations $\mathbf{A}_V$ of image modality, where $\mathbf{U}_V \in \mathbb{R}^{k_c \times k_1}$ is the projection matrix. The same remarks are applied to $\mathbf{P}_T$ and $\mathbf{U}_T$. We require each pair of samples from $\mathbf{A}_V$ and $\mathbf{A}_T$ can be projected to the same instance of $\mathbf{P}_V$ and $\mathbf{P}_T$ in space $\mathcal{P}$ (see the illustration in Fig. 2). To be more precise, for an instance $\mathbf{p} = \mathbf{U}_V \alpha_V$ (or $\mathbf{p} = \mathbf{U}_T \alpha_T$) in $\mathcal{P}$ which is projected from the sparse representation α_V (or α_T) of image (or text) modality, we can reconstruct the corresponding sparse representation of text (or image) modality $\alpha_T = \mathbf{U}_T^{-1}\mathbf{p}$ (or $\alpha_V = \mathbf{U}_V^{-1}\mathbf{p}$). Finally, the definition of $f(\mathbf{A}_V, \mathbf{A}_T)$ can be formulated as:

$$\begin{aligned} f(\mathbf{A}_V, \mathbf{A}_T) = & \gamma(\|\mathbf{A}_V - \mathbf{U}_V^{-1}\mathbf{P}_T\|_F^2 + \|\mathbf{A}_T - \mathbf{U}_T^{-1}\mathbf{P}_V\|_F^2) \\ & + \mu(\|\mathbf{U}_V^{-1}\|_F^2 + \|\mathbf{U}_T^{-1}\|_F^2), \quad (2) \end{aligned}$$

where the additionally regularized constraints on $\mathbf{U}_V^{-1}$ and $\mathbf{U}_T^{-1}$ ensure numerical stability and to avoid over-fitting. It can be learnt that the constraints in Eq. 2 exhibits the capabilities in recovering the sparse representations in one modality using data projected from the other, hence the relationship across different modalities can be efficiently incorporated in the sparse representations.

By substituting Eq. 2 into Eq. 1, we obtain the objective function for coupled dictionary learning procedure. Although the objective function is not jointly convex with respect to dictionaries $\{\mathbf{D}_V, \mathbf{D}_T\}$, sparse representations $\{\mathbf{A}_V, \mathbf{A}_T\}$ and projection matrices $\{\mathbf{U}_V, \mathbf{U}_T\}$, it is convex with respect to each of them when fixing the other variables. We can iteratively update them in an alternating manner. We outline the proposed iterative algorithm in Algorithm 1 and the detailed optimization is provided in the supplementary. And we will also release our code for public usage.

3.2. Coupled feature mapping

Let $\mathbf{Y} = [\mathbf{y}_1, \mathbf{y}_2, ..., \mathbf{y}_n] \in \mathbb{R}^{c \times n}$ be the keyword matrix for all images in $\mathcal{D}$. The coupled feature mapping procedure aims to learn two projection matrices $\mathbf{W}_V \in \mathbb{R}^{k_1 \times c}$ and $\mathbf{W}_T \in \mathbb{R}^{k_2 \times c}$, which map the sparse representations $\mathbf{A}_V$ and $\mathbf{A}_T$ of the two modalities into the common space defined by keywords. Note that the role of $\mathbf{W}_V$ and $\mathbf{W}_T$ is different from $\mathbf{U}_V$ and $\mathbf{U}_T$ of coupled dictionary learning procedure, as the former bridges the sparse representations to the keyword space, while the latter performs on the intermediate space $\mathcal{P}$.

In this work, we leverage the coupled linear regression method and develop two schemes (a simple scheme and an advanced scheme) for coupled feature mapping procedure, based on how the projection matrices are learnt. Specifically, for the simple scheme, we adopt the ridge regression method to minimize the errors of projecting the sparse representations of each modality to the keyword space. The generic minimization problem of the simple scheme is derived as:

$$\begin{aligned} \min_{\mathbf{W}_V, \mathbf{W}_T} & \|\mathbf{W}_V^\top \mathbf{A}_V - \mathbf{Y}\|_F^2 + \|\mathbf{W}_T^\top \mathbf{A}_T - \mathbf{Y}\|_F^2 \\ & + \lambda(\|\mathbf{W}_V\|_F^2 + \|\mathbf{W}_T\|_F^2). \quad (3) \end{aligned}$$

The analytical solutions of $\mathbf{W}_V$ and $\mathbf{W}_T$ can be derived as:

$$\begin{aligned} \mathbf{W}_V &= \mathbf{Y}\mathbf{A}_V^\top(\mathbf{A}_V\mathbf{A}_V^\top + \lambda I)^{-1} \\ \mathbf{W}_T &= \mathbf{Y}\mathbf{A}_T^\top(\mathbf{A}_T\mathbf{A}_T^\top + \lambda I)^{-1} \quad (4) \end{aligned}$$

For the advanced scheme, we follow the coupled feature mapping method in [13] to further incorporate coupled feature selection. Intuitively, relevant and discriminative features

from the sparse representations can be selected to enhance the relevance across different modalities. Therefore, for the advanced scheme, it has similar objective function as in [13]:

$$\min_{\mathbf{W}_V,\mathbf{W}_T} (\|\mathbf{W}_V^\top \mathbf{A}_V - \mathbf{Y}\|_F^2 + \|\mathbf{W}_T^\top \mathbf{A}_T - \mathbf{Y}\|_F^2)/2 \quad (5)$$
$$+ \lambda_1(\|\mathbf{W}_V\|_{21} + \|\mathbf{W}_T\|_{21}) + \lambda_2\|[\mathbf{W}_V^\top \mathbf{A}_V \mathbf{W}_T^\top \mathbf{A}_T]\|_*,$$

where the L_{21} norm $\|\cdot\|_{21}$ is used to select features from coupled feature space jointly, the trace norm $\|\cdot\|_*$ encodes the correlations of different modalities with a low-rank constraint. It is worth noting that our advanced scheme in Eq. 5 differs from [13] that coupled feature selection is performed on sparse representations $\mathbf{A}_V$ and $\mathbf{A}_T$ to leverage the merit from coupled dictionary learning, rather than on original features of both modalities in [13]. In practice, we use the iterative algorithm developed in [13] to optimize Eq. 5.

Algorithm 1 Iterative Algorithm for our proposed framework

Input: Image feature matrix $\mathbf{V}$ with associated text feature matrix $\mathbf{T}$, parameters σ, γ and μ.

1: Initialize $\{\mathbf{D}_V^0, \mathbf{D}_T^0\}$ and $\{\mathbf{A}_V^0, \mathbf{A}_T^0\}$ by [14], and $\{\mathbf{U}_V^0, \mathbf{U}_T^0\}$ as $\mathbf{I}$ for iteration $i = 0$.
2: Let $\mathbf{P}_V^0 \leftarrow \mathbf{U}_V^0 \mathbf{A}_V^0$ and $\mathbf{P}_T^0 \leftarrow \mathbf{U}_T^0 \mathbf{A}_T^0$.
3: **repeat**
4: Update $\mathbf{D}_V^{i+1}$, $\mathbf{D}_T^{i+1}$ with $\mathbf{A}_V^i$, $\mathbf{A}_T^i$ and $\mathbf{U}_V^i$, $\mathbf{A}_T^i$.
5: Update $\mathbf{A}_V^{i+1}$, $\mathbf{A}_T^{i+1}$ with $\mathbf{D}_V^{i+1}$, $\mathbf{D}_T^{i+1}$ and $\mathbf{U}_V^i$, $\mathbf{A}_T^i$.
6: Update $\mathbf{U}_V^{i+1}$, $\mathbf{U}_T^{i+1}$ with $\mathbf{D}_V^{i+1}$, $\mathbf{D}_T^{i+1}$ and $\mathbf{A}_V^{i+1}$, $\mathbf{A}_T^{i+1}$.
7: Update $\mathbf{P}_V^{i+1} \leftarrow \mathbf{U}_V^{i+1} \mathbf{A}_V^{i+1}$ and $\mathbf{P}_T^{i+1} \leftarrow \mathbf{U}_T^{i+1} \mathbf{A}_T^{i+1}$.
8: Set $i = i + 1$.
9: **until** Objective function of Eq.1 converges.

Output: Dictionaries $\{\mathbf{D}_V, \mathbf{D}_T\}$, sparse representations $\{\mathbf{A}_V, \mathbf{A}_T\}$.

Input: Sparse representations $\{\mathbf{A}_V, \mathbf{A}_T\}$, keywords matrix $\mathbf{Y}$, parameters λ (or $\{\lambda_1, \lambda_2\}$).

10: Compute $\{\mathbf{W}_V, \mathbf{W}_T\}$ according to Eq. 4 or solutions of Eq. 5.

Output: Projection matrices $\{\mathbf{W}_V, \mathbf{W}_T\}$.

3.3. Test phase for cross-modal retrieval

Algorithm 1 summarizes the iterative algorithms used in the above two procedures of the training stage. Once the training stage is complete, we can obtain the dictionaries $\{\mathbf{D}_V, \mathbf{D}_T\}$, sparse representations $\{\mathbf{A}_V, \mathbf{A}_T\}$, projection matrices $\{\mathbf{W}_V, \mathbf{W}_T\}$ of two modalities. In the testing phase, given a test image $\hat{I} = (\hat{\mathbf{v}}, \hat{\mathbf{t}})$, we first generate the sparse representations $\hat{\mathbf{A}}_{\hat{v}}$ and $\hat{\mathbf{A}}_{\hat{t}}$ based on learnt dictionaries $\mathbf{D}_V$ and $\mathbf{D}_T$, respectively. Then we can project $\hat{\mathbf{A}}_{\hat{v}}$ and $\hat{\mathbf{A}}_{\hat{t}}$ into the keyword space through projection matrices $\mathbf{W}_V$ and $\mathbf{W}_T$, respectively. To perform cross-modal retrieval, we take one modal data (i.e., image modality) of $\hat{I}$ as the query to retrieval the other modality (i.e., text modality).

4. EXPERIMENTAL RESULTS

4.1. Experimental setting

We apply the proposed framework to two cross-modal retrieval tasks: *Text2Img* and *Img2Text*. Given an image (or text) query, the goal is to find the nearest neighbors from text (or image) database. We compare the two proposed schemes (simple and advanced) with several related methods on two publicly available datasets: Pascal Voc and Wiki dataset , in which each image is assigned with only one keyword. In addition, we further investigate a more specific task called *Img2Keyword*. The goal of *Img2Keyword* to predict the results from the keyword space rather than retrieving a sample from the text modality for a given image. It can be considered as the traditional image annotation task and here we would like to use it to evaluate the efficiency of the proposed feature mapping into the common subspace defined by keywords. For this task, we adopt a challenging dataset, MIRFlickr-25K , in which each image is assigned with at least one keyword. Table 1 lists some of the general statistics of the three datasets used in this work.

Table 1. General statistics of three datasets used in this work.

Dataset	Images	Keywords	Image feature	Text feature
Pascal Voc	2808, 2841	20	512-dim Gist	399-dim word frequency
Wiki	1300, 1566	10	128-dim SIFT	10-dim LDA features
MIRFilckr-25K	12500, 12500	38	7500-dim multiple features	457-dim word frequency

To evaluate the performance of the proposed schemes, we use the standard measure of mean average precision (MAP) in [8] for *Img2Text* and *Text2Img* tasks. To compute MAP, we first evaluate the average precision (AP) of the retrieval result of each query, and then average the AP values over all queries in the query set. Moreover, to evaluate the *Img2Keyword* task, we adopt the standard measures of image annotation task: average precision per keyword (P), average recall per keyword (R) and $F1$ measure ($F1 = 2 \times \frac{P \times R}{P+R}$). The values of P and R are computed by predicting the top five keywords for each test image on MIRFlickr-25K dataset. Notable that for all the measures, larger numerical value indicates better performance. In the training stage, we empirically set the parameters σ, γ and μ to 0.01, 0.0001 and 0.001 for the coupled dictionary learning procedure; and the parameters λ, λ_1 and λ_2 to 0.01, 0.1 and 0.001 for the feature mapping procedure, respectively. In testing phase, the cosine distance is adopt to measure the similarity of features and select the matches.

4.2. Results on Pascal Voc and Wiki datasets

We first compare the two proposed schemes with common subspace learning methods PLS [11], CCA [10], GMMFA [12], GMLDA [12] and LCFS [13] and report the results for *Img2Text* and *Text2Img* tasks. Using the settings of LCFS

method, we perform Principal Component Analysis (PCA) on the original features of Pascal Voc dataset to remove redundant features for PLS, CCA, GMMFA, GMLDA methods while preserving the original features for LCFS method and the proposed two schemes. Since the Wiki dataset has low dimensional features, we directly use theses features.

Table 2 shows the results of different methods in terms of MAP on Pascal Voc dataset. We can see that our proposed simple scheme significantly outperforms the previous methods. It shows that our coupled dictionary learning algorithm has the advantage of outputting sparse representation that preserves the relationship among different modalities. Moreover, the advanced scheme makes further improvement over the simple one, showing that coupled feature selection can be done in the sparse representations and benefits the learned common space for feature mapping.

Table 2. MAP scores of different methods on Pascal Voc dataset.

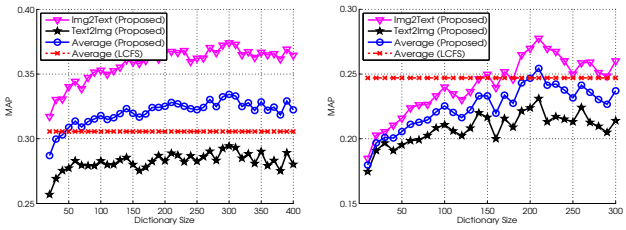
Method	Img2Text	Text2Img	Average
PCA+PLS	0.2757	0.1997	0.2377
PCA+CCA	0.2655	0.2215	0.2435
PCA+GMMFA	0.3090	0.2308	0.2699
PCA+GMLDA	0.2418	0.2038	0.2228
LCFS	0.3438	0.2674	0.3056
Proposed (Simple)	0.3262	0.2997	0.3129
Proposed (Advanced)	0.3741	0.2944	0.3342

Table 3 shows the MAP scores of different approaches on the Wiki dataset. On average, our proposed schemes achieves higher MAP scores than the other methods but perform worse than some methods on *Img2Text* task. As discussed in [13], it is challenging to improve on this dataset due to the low dimensions of image and text features. Nevertheless, our proposed schemes consistently outperform previous methods especially on *Text2Img* task. To further understand the reason, we change the dictionary size of the coupled dictionary learning step in the advanced scheme and compare the performance. Figure 3 shows the MAP scores with different dictionary size on the two datasets. We can see that larger dictionary size generally has better capability for sparse representation and the best MAP scores on the two datasets are achieved with dictionary size 300 and 210, respectively. Therefore, the limitation of low dimensional (10-dim) text features (see Table 1) of the Wiki dataset can be tackled by our coupled dictionary procedure where we use more efficient high dimensional (210-dim) sparse features, benefitting the coupled feature selection in the feature mapping procedure. However, the decay on *Img2Text* indicates that the sparse representations of different modalities may need to be further balanced.

Specifically, for *Text2Img* task, Fig. 4 shows two examples of text queries and the top five images retrieved using the advanced scheme. In each example, the text query and its

Table 3. MAP scores of different methods on Wiki dataset.

Method	Img2Text	Text2Img	Average
PLS	0.2402	0.1663	0.2032
CCA	0.2549	0.1846	0.2198
GMMFA	0.2750	0.2139	0.2445
GMLDA	0.2751	0.2098	0.2425
LCFS	0.2798	0.2141	0.2470
Proposed (Simple)	0.2717	0.2233	0.2475
Proposed (Advanced)	0.2776	0.2311	0.2544



(a) MAP on Pascal Voc (b) MAP on Wiki

Fig. 3. The MAP with different dictionary size of the proposed scheme (advanced) on the Pascal Voc dataset (left) and Wiki dataset (right). Note that the best MAP score (Average) of LCFS method is also provided for guides in red dotted line.

corresponding image are shown at the left, and the top five images retrieved are listed in the following columns. It can be observed that our advanced scheme finds the closet matches of the image modality at the semantic level for both text queries. And the retrieved images are all belonging to the same keyword of the text queries ("music" at the top, "warfare" at the bottom).

4.3. Results on MIRFlickr-25K dataset

We then perform comparison for the *Img2Keyword* task. To make comprehensive comparison, we also take into account several state-of-the-art image annotation methods, which can also be evaluated for the *Img2Keyword* task. These methods include: (1) JEC [4] , Tagprop [5] and Fasttag [15], which only use image modality for learning model; (2) Multi-kernel SVMs [2], which applies different kernel functions to train SVM classifiers for image and text modalities; (3) Kernel CCA (KCCA) [6] and LCFS [13], which are common subspace learning methods. Since KCCA does not directly map image modal data into the keyword space, we use it for nearest neighbor selection then combine it with the nearest neighbor based tag propagation scheme Tagprop (this has been reported with promising result in [6]). For our proposed scheme, we set dictionary size as 350 for coupled dictionary learning.

We report the overall performance on *Img2Keyword* task in Table 4 and more visualized examples in the supplementary. We observe that: 1) using an additional text modality

Fig. 4. Two examples of the *Text2Img* task obtained by our proposed advanced scheme on Wiki dataset.

Table 4. Comparison of *Img2Keyword* in terms of P, R and F1 measure for the proposed schemes and related methods on MIRFlickr-25K dataset.

Method	P	R	F1
JEC	0.3290	0.1733	0.2270
Tagprop	0.4521	0.3021	0.3622
Fasttag	0.4498	0.3637	0.4021
Multi-kernel SVMs	0.5162	0.3658	0.4282
KCCA+Tagprop	0.5469	0.3542	0.4299
LCFS	0.3253	0.3122	0.3186
Proposed (Simple)	0.4933	0.3445	0.4056
Proposed (Advanced)	0.5890	0.3779	0.4604

improves the accuracy of *Img2Keyword* task than only using the image modality; 2) both of our proposed schemes outperform the subspace learning method LCFS, indicating that the coupled dictionary learning procedure in our proposed framework is efficient to handle the diversity of different modalities and the learned sparse representations is more powerful than the original features for subspace learning; 3) our advanced scheme achieves the highest performance and generally outperforms the state-of-the-art method KCCA+Tagprop, showing that the learned sparse representations are powerful for subspace learning and coupled feature selection is crucial in enhancing the relationships across different modalities.

5. CONCLUSION

In this paper, we propose a novel framework solving the cross-modal retrieval problem. And we develop an efficient iterative algorithm for solving the optimization problem in the proposed framework. Experiments on various cross-modal retrieval tasks give significantly better performance over related state-of-the-art methods, showing the superiority of the proposed framework. In the future, we will explore the potentiality of the proposed framework in fill in missing modalities given the observed one, to deal with the issue of absence of some modalities.

6. REFERENCES

[1] Vicente Ordonez, Girish Kulkarni, and Tamara L. Berg, "Im2text: Describing images using 1 million captioned photographs," in *NIPS*, 2011.

[2] J. Verbeek, M. Guillaumin, T. Mensink, and C. Schmid, "Image annotation with tagprop on the mirflickr set," in *ACM MIR*, 2010, MIR '10, pp. 537–546.

[3] Y. Verma and CV. Jawahar, "Im2text and text2im: Associating images and texts for cross-modal retrieval," in *BMVC*, 2014.

[4] A. Makadia, V. Pavlovic, and S. Kumar, "A new baseline for image annotation," in *ECCV*, 2008.

[5] M. Guillaumin, T. Mensink, J. Verbeek, and C. Schmid, "Tagprop: Discriminative metric learning in nearest neighbor models for image auto-annotation," in *CVPR*, 2009, pp. 309 –316.

[6] L. Ballan, T. Uricchio, L. Seidenari, and A. Del Bimbo, "A cross-media model for automatic image annotation," in *ICMR*, 2014.

[7] R. Datta, D. Joshi, J. Li, and J. Wang, "Image retrieval: Ideas, influences, and trends of the new age," *ACM Comput. Surv.*, vol. 40, pp. 1–60, 2008.

[8] N. Rasiwasia, J. Costa Pereira, E. Coviello, G. Doyle, G. Lanckriet, R. Levy, and N. Vasconcelos, "A new approach to cross-modal multimedia retrieval," in *ACM MM*, 2010.

[9] Y. Gong, Q. Ke, M. Isard, and S. Lazebnik, "A multi-view embedding space for modeling internet images, tags, and their semantics," *IJCV*, vol. 106, pp. 210–233, 2014.

[10] S. Hwang and K. Grauman, "Learning the relative importance of objects from tagged images for retrieval and cross-modal search," *IJCV*, vol. 100, pp. 134–153, 2012.

[11] Y. Chen, L. Wang, W. Wang, and Z. Zhang, "Continuum regression for cross-modal multimedia retrieval," in *ICIP*, 2012.

[12] A. Sharma, A. Kumar, H. Daume, and D. W. Jacobs, "Generalized multiview analysis: a discriminative latent space," in *CVPR*, 2012, pp. 2160–2167.

[13] K. Wang, R. He, W. Wang, L. Wang, and T. Tan, "Learning coupled feature spaces for cross-modal matching," in *ICCV*, 2013, pp. 2088–2095.

[14] D. Huang and Y. Wang, "Coupled dictionary and feature space learning with applications to cross-domain image synthesis and recognition," in *ICCV*, 2013, pp. 2496–2503.

[15] M. Chen, A. Zheng, and K. Weinberger, "Fast image tagging," in *ICML*, 2013, pp. 1274–1282.